

ECON 381: Homework 1

Fall 2026

Due: Thursday, September 3, 2026, at 11:59 p.m. Eastern Time

Submit electronically through the Canvas **Assignments** section.

Purpose

In this assignment, you will write and run a Stata do-file that imports a dataset, produces descriptive statistics, creates an indicator variable, compares groups, makes a scatter plot with a fitted line, and records the complete session in a log file. Any requested interpretation or answer must be written as a Stata comment directly in the do-file; no separate written-answer document is required.

Files and submission requirements

Download `caschool.dta` from Canvas and place it in a working folder on your computer. Submit all three of the following files on Canvas:

- your Stata do-file: `ECON381_Stata_LastName.do`;
- your Stata log file: `ECON381_Stata_LastName.log`; and
- the scatter plot exported as `ECON381_Stata_LastName_Graph.pdf`.

Your do-file must run from beginning to end without errors. Use comments to identify each numbered task. Provide every requested answer or interpretation as a Stata comment immediately after the relevant command. Use a relative file path or a clearly marked working-directory command so that the file can be adapted to another computer. The log must show the commands, output, and results produced by your do-file.

You may submit individually or in a group of two, as allowed by the syllabus. If you work with a partner, **both members must submit identical work**, list both names at the top of every submitted file, and follow the submission instructions given in this document.

Assignment tasks

Write one Stata do-file that completes all of the following tasks in order.

1. **Start the log and load the data.** Use Stata's `log using` command near the beginning of your do-file so that all subsequent commands and output are recorded. Then load `caschool.dta` into Stata. Include the appropriate option if data already in memory must be cleared.
2. **Summarize average test scores.** The variable `testscr` contains each district's average test score. Report its mean, standard deviation, minimum, and maximum. In a Stata comment in your do-file, label and report each statistic clearly.
3. **Classify districts by income.** The variable `avginc` contains average district income. First obtain and store the sample median of `avginc`. Then generate a new variable named `inc_indicator` that equals 1 for a high-income district (income strictly above the sample median) and 0 for a low-income

district (income strictly below the sample median). Add value labels so the two groups are easy to identify. If any district has income exactly equal to the median, leave `inc_indicator` missing and state how many such observations there are.

4. **Compare test scores across income groups.** Obtain the mean of `testscr` separately for high-income and low-income districts. In a Stata comment, report both means and briefly describe what you find: which group has the higher average test score, and approximately how large is the difference?
5. **Summarize districts in Los Angeles County.** Use the `county` variable to restrict the calculation to observations for Los Angeles County. Compute the mean of `testscr` and the mean of `avginc` for those districts. In a Stata comment, report both means clearly. Make sure your string comparison uses the county name exactly as it appears in the dataset.
6. **Graph test scores and income.** Create a scatter plot with `testscr` on the vertical axis and `avginc` on the horizontal axis. On the same graph, add a linear fitted line. Give the graph an informative title and axis labels, and export it using the required filename. In a Stata comment, state whether the fitted line slopes upward or downward and explain what this indicates about the association between district income and average test scores.
7. **Finish the log.** Use `log close` at the end of the do-file. Before submitting, open the log and confirm that it contains the commands, comments, and output for Tasks 1–6 and that the do-file ran without errors.

Late submission policy

Late assignments are accepted with the following reduction to the earned score:

- at most 3 hours late: 80% of the earned score;
- more than 3 and at most 6 hours late: 60% of the earned score;
- more than 6 and at most 9 hours late: 40% of the earned score;
- more than 9 and at most 12 hours late: 20% of the earned score; and
- more than 12 hours after the deadline: a score of zero.

Assignments are made available sufficiently early. Because students can plan ahead and submit early, illness, university events, and other documented circumstances do not ordinarily excuse a late submission. Please plan ahead and submit early when you anticipate a potential conflict. The lowest assignment grade will be dropped, as described in the Fall 2026 syllabus.

Before you submit

Confirm that (i) all required files open correctly, (ii) all group members' names appear in the files, if applicable, (iii) every requested answer appears as a Stata comment in the do-file, (iv) the graph is readable, and (v) the log records the entire Stata session. Missing code, output, comments, or required files may prevent the corresponding work from being graded.